

7-2c How a Higher Price Raises Producer Surplus

You will not be surprised to hear that sellers always want to receive a higher price for the goods they sell. But how much does sellers' well-being rise in response to a higher price? The concept of producer surplus offers a precise answer to this question.

Figure 6 shows a typical upward-sloping supply curve that would arise in a market with many sellers. Although this supply curve differs in shape from the previous figure, we measure producer surplus in the same way: Producer surplus is the area below the price and above the supply curve. In panel (a), the price is P_1 and producer surplus is the area of triangle ABC.

Figure 6

How Price Affects Producer Surplus

In panel (a), the price is P_1 , the quantity supplied is Q_1 , and producer surplus equals the area of the triangle ABC. When the price rises from P_1 to P_2 , as in panel (b), the quantity supplied rises from Q_1 to Q_2 and the producer surplus rises to the area of the triangle ADF. The increase in producer surplus (area BCFD) occurs in part because existing producers now receive more (area BCED) and in part because new producers enter the market at the higher price (area CEF).

Panel (b) shows what happens when the price rises from P_1 to P_2 . Producer surplus now equals area ADF. This increase in producer surplus has two parts. First, those sellers who were already selling Q_1 of the good at the lower price P_1 are better off because they now get more for what they sell. The increase in producer surplus for existing sellers equals the area of the rectangle BCED. Second, some new sellers enter the market because they are

willing to produce the good at the higher price, resulting in an increase in the quantity supplied from Q_1 to Q_2 . The producer surplus of these newcomers is the area of the triangle CEF.

As this analysis shows, we use producer surplus to measure the well-being of sellers in much the same way as we use consumer surplus to measure the well-being of buyers. Because these two measures of economic welfare are so similar, it is natural to consider them together. Indeed, that is exactly what we do in the next section.

QuickQuiz

4. Diego, Emi, and Finn are available to work as tutors for the semester. The opportunity cost of tutoring is \$100 for Diego, \$200 for Emi, and \$400 for Finn. The university is hiring tutors at a price of \$300. Producer surplus equals
- a. \$100.
 - b. \$200.
 - c. \$300.
 - d. \$400.
5. Gavin has been working full-time as a gardener for \$300 a week. When the market price of gardeners rises to \$400, Hector becomes a gardener as well. How much does producer surplus rise as a result of this price increase?
- a. by less than \$100
 - b. between \$100 and \$200
 - c. between \$200 and \$300
 - d. by more than \$300
6. The supply curve for a product is $Q^s = 2P$, and the market price is \$10. What is producer surplus? (*Hint: Graph the supply curve and recall the formula for the area of a triangle.*)
- a. \$5
 - b. \$20

c. **\$100**

d. **\$200**

Chapter 7: Consumers, Producers, and the Efficiency of Markets: 7-2c How a Higher Price Raises Producer Surplus

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